CS 138 Homework 1: Multi-Armed Bandits in Non-Stationary Settings

For this homework, you will work on the tasks listed below. Your submission should include a report, about 5-6 pages, using the standard LaTeX overleaf template:

https://www.overleaf.com/read/wvjzcxdkxtp

Example reports are available on the course website. The report should be in PDF format. Note that the examples are of previous semester assignments.

Part 1: Non-stationary MABs using a random walk

For this part of the homework, implement and conduct experiments as described in Exercise 2.5 of the book. Your results should include plots that show reward over time, as well as % optimal action selected, over multiple conditions as specified in the exercise. Feel free to experiment with different initialization functions, different rates of exploration, etc.

Part 2: Soft-max exploration

Using the implementation in Part 1, implement a MAB agent that uses soft-max action selection and compare the results with e-greedy action selection with decaying epsilon over time and for soft-max, decaying temperature over time. Show plots that compare the two different action selection methods. Discuss the results and make some hypotheses as to why they appear the way they are.

Part 3: Other types of non-stationarity

Design two additional types of non-stationarity that are different from the random walk described in Ex. 2.5. Be creative! In 2.5, the change in the environment is gradual but it doesn't need to be. For example, one additional type of non-stationarity is if two of the arms “flip” and now each one has the average payout of the other. Conduct experiments similar to those in Part 1 and 2 and compare various approaches for action selection and evaluate over different hyperparameter values.

Extra Credit

For extra credit you can do 2 of the non-programming exercises in Chapter 2. Describe in detail how you approached the problem and what solution you came up with. Usually 1-2 paragraph plus some math or notation is enough per exercise.

Alternatively, solve the other programming exercise from Chapter 2.

Or do both for extra extra credit. **If you do anything for extra credit, make sure your paper ends with a section called "Extra Credit".**

General Guidelines (PLEASE READ!):

Your paper should be detailed enough that anyone reading it, with access to the textbook, can **replicate** your results. This is the gold standard for science.

Your code should be object-oriented, at a minimum, you will need a class for the MAB agent and a class for the MAB environment with clearly defined interface between the two so that there is no information leakage from the environment to the class to the agent class. We will discuss such interfaces in class. You can use any language of your choice that supports OO-programming, but we recommend Python.

Your submission **MUST** include:

1. PDF of the paper (usually 5-6 pages but you can go longer if you need to explain further details so your work is reproducible or have additional interesting result plots to show)
2. All code (colab/jupyter notebooks are fine)
3. A log of the prompts and results when or if using gen-AI to generate code
4. A README.txt file that provides instructions for how someone could run your code.

Policy on using gen-AI:

You are allowed to use tools such as Copilot, Claude, etc. as you program your solutions as well as for generating the code to make plots (I recommend using pyplot if using python). However, the writing of the report should be done on your own. Of course, you can use AI tools for grammar check, spell check, and small stylistic edits. You must log your prompts and their outputs when gen-AI is used.

Policy on collaboration:

You are allowed to collaborate with your classmates as follows:

1. You can discuss at the conceptual level all aspects of the homework.
2. You can discuss experiment designs for each part.
3. You can play with gen-AI with various prompts and discuss the results.
4. You can review your buddies' papers and provide feedback

What you need to do on your own:

1. Run your experiments and plot the results.
2. Write the paper
3. Implement/code your own solutions

If you collaborated with anyone, add a note at the end, state whom you collaborated with and write a few sentences about what the collaboration consisted of.